

Education

University of Connecticut

PhD Candidate in Statistics, Department of Statistics

- Courses: *Design of Experiments, Statistical Inference, Linear Models*

Sep 2024 – Present

Storrs, CT, U.S.

Renmin University of China

Master of Science in Applied Statistics, Institute of Statistics and Big Data

- Courses: *Clinical Trial Statistical Methodology, Causal Inference, Survival Analysis*

Sep 2022 – Jun 2024

Beijing, China

Xi'an Jiaotong University

Bachelor of Economics in Finance, School of Economics and Finance

- Courses: *Probability Theory, Mathematical Statistics, Econometrics*

Sep 2018 – Jun 2022

Xi'an, China

Work Experience

Astellas Pharma

Biostatistician Intern

May 2026 – Aug 2026

Remote

Topic: Closed-Form Evaluation of Correlated PFS and OS Group Sequential Designs

- Developed an event-driven group sequential design for correlated progression-free survival (PFS) and overall survival (OS) under the Fleischer model, supporting efficacy monitoring, futility stopping, and piecewise-uniform accrual.
- Derived a closed-form joint correlation matrix for sequential PFS and OS log-rank statistics, established a multivariate-normal asymptotic approximation, and calculated boundaries in three scales (z-score, p-value, and hazard ratio), marginal power, and gatekeeping joint power.
- Built an R package and integrated R Shiny application that automate trial-design specification, closed-form analysis, visualization, and Monte Carlo simulation-based validation, enabling efficient and reproducible design evaluation.

Research Experience

Department of Statistics, UConn

Dissertation Research, Advised by Prof. Haiying Wang and Prof. Jun Yan

Sep 2025 – Present

Storrs, CT, U.S.

Topic: Rerandomization Methods for Experimental Design

- Conducting a comprehensive review of rerandomization methodologies, synthesizing core theoretical developments, and identifying open research questions in the field.
- Investigating the adverse impact of high-dimensional covariates on rerandomization, including strategies to mitigate high-dimensional imbalance and analysis of the associated statistical properties.
- Developing principled variable selection strategies under rerandomization to identify influential covariates while maintaining design validity and statistical efficiency.
- Building a high-performance R package for rerandomization-based experimental design and analysis, incorporating recent theoretical advances with scalable and user-friendly implementations.

Department of Public Health Sciences, UConn Health

Research Assistant

Sep 2025 – Dec 2025

Farmington, CT, U.S.

Topic: Longitudinal Analysis of Depression Outcomes in Large-Scale Medicare Claims Data

- Processed multi-billion Medicare claims datasets using high-performance big-data tools in R, including `fst` and `data.table`, enabling efficient extraction and linkage of depression patient cohorts across multiple years.
- Constructed a longitudinal cohort of Medicare beneficiaries with depression by integrating claims histories, chronic condition indicators, and demographic covariates.
- Applied survival analysis methods to identify key clinical and demographic factors associated with the onset and progression of depression among Medicare patients.

Center for Population Health, UConn Health

Research Assistant

Jan 2025 – Jun 2025

Farmington, CT, U.S.

Topic: Survival and Machine Learning Analysis of Suicide Risk Using Clinical Data

- Conducted data mining on patient demographics and clinical records, employing logistic regression and Weibull Cox proportional hazards models to identify key factors associated with suicide rates.
- Implemented survival analysis techniques to address missing data challenges and assess time-to-event outcomes related to suicide risk.

- Employed multiple machine learning techniques, including Random Forest, SVM, and Neural Networks, to predict suicide rates and improve forecasting accuracy.

Connecticut Children's Medical Center

Research Assistant

Aug 2024 – Jun 2025

Hartford, CT, U.S.

Topic: Time-Series Forecasting of Emergency Department Patient Volume

- Preprocessed and visualized emergency department census data from Connecticut Children's Medical Center, using heatmap calendar plots to reveal key periodic patterns and temporal trends in patient flow.
- Collected external covariates (temperature, air pollution, water quality) from public APIs, integrated them with census data, and performed data merging for downstream modeling and analysis.
- Utilized Fourier series to construct a harmonic dynamic regression model to estimate different periodic cycles, combined with an ARIMAX framework to model non-periodic short-term variations, achieving effective forecasting performance.

Institute of Geriatric Medicine, Beijing Hospital

Research Assistant

Oct 2022 – May 2023

Beijing, China

Topic: Change-Point Analysis of Physiological in Aging Populations

- Streamlined and preprocessed population data from Beijing Hospital, and utilized line plots and cubic spline regression plots to identify significant change points in various physiological function variables.
- Estimated knots in linear spline models using the Newton-Raphson algorithm, specifically tailored to capture critical shifts related to the aging process.
- Developed advanced plotting functions to visually validate results and engaged in thorough consultations with hospital doctors, ensuring a robust integration of statistical findings with clinical experiences.

Institute of Statistics and Big Data, Renmin University of China

Dissertation Research, Advised by Prof. Yang Liu

Mar 2023 – Jun 2024

Beijing, China

Topic: Covariate-Adaptive Randomization for Multi-Arm Experiment with Unequal Allocations

- Developed improved covariate-adaptive randomization procedures for multi-arm clinical trials, proposing novel randomization schemes that enhance covariate balance while preserving randomization integrity.
- Studied theoretical properties of CAR procedures by deriving asymptotic distributions of covariate imbalances under general multi-arm designs with unequal allocation ratios.

Teaching Experience

Institute of Statistics and Big Data, Renmin University of China

Teaching Assistant for Course Clinical Trial Methodology

Sep 2023 – Jan 2024

Beijing, China

Publications

- Xingjian Ma, Robert H. Aseltine, Tze Chiam, Richelle deMay, Yingfa Xie, and Jun Yan. Forecasting Emergency Department Census: Leveraging Periodic Patterns and External Covariates. (*Major Revision*)
- Xie, Y., Aseltine, R. H., Chiam, T., deMayo, R., Huang, Y., Ma, X., & Yan, J. (2025). Modeling Emergency Department Volume-at-Risk Using a Discrete Extreme Value Approach. *The Journal of the Royal Statistical Society: Series C*. (*Major Revision*)
- Xingjian Ma and Yang Liu. Balancing Observed and Unobserved Covariates with Multi-arm Covariate-Adaptive Randomization for Unequal Allocations. (*In preparation*)

Projects

Parkinson's Disease Progression Prediction | Python (PyTorch)

May 2023 – Jun 2023

- Engineered time-aware clinical features from high-dimensional proteomic data, reducing 1,200+ variables to 25 informative predictors, enhancing model interpretability and performance.
- Developed a hybrid model combining LightGBM (multi-class classification) and a neural network and applied strict time-series cross-validation to ensure robust generalization on future patient visits.

Covariate-Adjusted Randomization in Group Sequential Design | R, Rcpp

Feb 2023 – Dec 2023

- Incorporated covariates in a multivariate regression analysis to aid in estimating the variance of treatment effects, further optimizing the efficiency of interim analysis and reducing the expected sample size.

rerand: R package for Rerandomization | R, Rcpp

Oct 2025 – Dec 2025

- Developed an R package **rerand** to implement rerandomization procedures in experimental design, allowing users to specify covariates and balance criteria for treatment assignment.

- Implemented Rcpp functions to optimize computational efficiency for large-scale rerandomization simulations, achieving a 5x speedup over pure R implementations.

Deep Reinforcement Learning | *Python (PyTorch)*

Oct 2022 – Dec 2022

- Developed AI-driven algorithms employing CNN architectures (e.g., AlexNet, VGGNet, ResNet) for enhanced interactive game image analysis, incorporating data augmentation and transfer learning techniques to boost model robustness.
- Utilized reinforcement learning strategies, specifically the Q-learning algorithm, to continuously train and refine the model based on dynamic game scenarios, resulting in improved decision-making and adaptability.

Honors & Awards

- | | |
|---|-----------------|
| • Outstanding Graduate of Renmin University of China | Apr 2024 |
| • Graduate Excellent First Class Scholarship, Renmin University of China | Oct 2023 |
| • Honorary Title of Outstanding Student Cadre, Renmin University of China | Oct 2023 |

Technical Skills

Programming: R (dplyr, tidyverse, ggplot2); Python (Numpy, Pandas, PyTorch); Rcpp; SAS; Git; Linux
Languages: Chinese(Native); English(Fluent - TOEFL: 100 (R28 L26 S22 W24))